

ERAP1 x axial spondyloarthritis

BOUNDED RECENT EVIDENCE ADDENDUM | 5 October 2026

The assessment incorporates a bounded addendum of recent primary studies from 2020-2026 for which sufficient methods and results were accessible for AXIS scientific curation.

CURRENT POSITION

Investigate context-specific ERAP1 modulation. Therapeutic benefit from ERAP1 inhibition is not established by this bounded evidence scope.

Decision: resolve the critical uncertainty first. No canonical DecisionState v2 has been created.

What the usable evidence changes

Recent accessible studies strengthen ERAP1-AS biological rationale, not validated therapeutic benefit. [CM01]

Full allotype, HLA/ERAP2 context and peptide substrate must be specified; a protective allotype is not universally inactive. [CM02]

Experimental benefit is endpoint-dependent: germline loss reduced rat arthritis incidence but worsened colon histology. [CM03]

AS is not every axSpA phenotype. Patient-genotype studies and disease models must not be silently generalized to every patient.

Additional potentially material literature was identified but not adjudicated where full methodological information was unavailable.

Independent scientific review pending.

Evidence strengthened. Attribution still open.

Recent accessible evidence	Decision relevance
Human genetics, PBMC and expanded CD8 responses	Stronger target/HLA context, not demonstrated drug benefit [CM01/05]
Substrate-sensitive recombinant and cellular allotype studies	Full sequence/context required; no universal low-activity label [CM02]
Divergent genetic-loss and inhibitor/KO phenotypes	Inhibition, lifelong loss and allosteric modulation differ [CM03/06]
New chemical series and regulatory-site complexes	More assay-specific tools; no intact-cell occupancy inferred [CM04]
Sparse ERAP2/IRAP counter-screen	Censored bounds retained; no non-comparable ratio [CM08]

Recent evidence addendum

10 primary studies integrated in the addendum; 21 atomic assertions and 10 impact assessments remain pending_review. This is not expert acceptance or a comprehensive review.

Four insufficient-access candidates remain outside assertion extraction. Corilagin 2025 is potentially material but its interaction/phenotype report cannot adjudicate intact-cell engagement without methods.

Clinical development, not efficacy

EAST1 (NCT07047703), an ERAP1-directed axSpA phase I/II trial, was registered with no posted results in the consulted record. No AS/axSpA efficacy results were identified in the searched sources. [REG01]

Chemical Learning: MODEL NOT BUILT in the unchanged indexed dataset. No new standardized measurements imported; recent-literature modelling eligibility was not assessed. [BASE01]

The next discriminating experiment

SAME PRIORITY, REFINED SPECIFICATION

Keep the matched-genotype engagement experiment; sharpen allotype, substrate, exposure and intracellular-versus-surface controls. [CM07]

Measure engagement and phenotype together

Use matched WT/null/rescue backgrounds, vehicle, dose/time, viability and interference controls. Resolve tested identity/form and validate a proximal cellular assay first. Specify HLA-B27 subtype, complete ERAP1 allotype and ERAP2 status. Distinguish intracellular folding/UPR from surface free-heavy-chain effects. [CM07]

Interpretable result	Scientific consequence
Engagement + dependency + rescue	Strengthens compound/context-specific ERAP1 attribution; translation and broader selectivity still need testing.
Phenotype persists without ERAP1	Weakens this compound's ERAP1 explanation, not the target universally.
Toxicity or invalid assay/genetic model	No mechanism conclusion; validate controls and repeat.

What remains unresolved

Relevant-exposure engagement and chemical-genetic dependency remain the discriminating gap for baseline phenotype-producing compounds. [CM06]

On-target, off-target, indirect and context-dependent explanations remain viable. New biology does not automatically select an inhibition strategy.

Bounded addendum does not replace the earlier failed exhaustive refresh: INSUFFICIENT REFRESH COMPLETENESS TO ASSESS. Full traceability and non-adjudicated reports are separated in the full brief.

Independent scientific review pending.